


```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Create a synthetic grayscale image
image = np.tile(np.arange(256, dtype=np.uint8), (256,1))

# Quantization function
def quantize(img, levels):
    step = 256 // levels
    return (img // step) * step

q256 = image
q64 = quantize(image, 64)
q16 = quantize(image, 16)
q8 = quantize(image, 8)
q4 = quantize(image, 4)
q2 = quantize(image, 2)
```